System Architecture & Application Workflow Specification

ShikshaIQ is an enterprise-grade adaptive learning platform engineered for Indian school education (NCERT Class 7 Mathematics: Fractions Chapter 2). In typical classrooms where one educator instructs 40 to 60 students, foundational deficits—such as Least Common Multiples (LCM) or Equivalent Fractions—often go undetected. ShikshaIQ resolves this bottleneck by integrating a **transparent deterministic knowledge graph** with an **isolated Socratic AI companion**, backed by real-time classroom telemetry and offline-first IndexedDB resilience.

NEXT.JS 16.4 APP ROUTER
DETERMINISTIC MASTERY ENGINE
SUPABASE POSTGRESQL + RLS
SOCRATIC AI WITH ZERO KEY LEAK
OFFLINE-FIRST PWA ENGINE

DISCOVERED TECHNOLOGY STACK & SYSTEM COMPONENTS

VERIFIED SOURCE BASE

System Layer	Technology / Framework	Architectural Function in ShikshaIQ
Frontend & UI	Next.js 16.4.3 (App Router), React 19.3, Tailwind CSS v4, Lucide Icons	Server Components, Turbopack bundling, minimal monochrome Apple-style UI, sub-second route transitions.
Edge Security	Next.js Edge Middleware (`middleware.ts`), HTTP-only SameSite Cookies	Enforces RBAC session validation via `@supabase/ssr`; routes unapproved teachers to `/pending-approval`.
Adaptive Engine	Deterministic DAG Engine (`lib/adaptive/`), TypeScript 5 (Strict)	Calculates score ratios without double-counting; identifies prerequisite bottlenecks via DFS graph traversal.
AI / ML Service	Google Gemini / OpenAI SDK Abstraction (`lib/ai/`)	Streaming Socratic companion with Indian cultural analogies (rotis, dosas, cricket). Answer keys strictly excluded.
Database & Auth	Supabase PostgreSQL (Managed Postgres 15+), Supabase Auth	ACID relational integrity, Row Level Security (RLS) policies on every table, append-only immutable audit logs.
Real-Time Telemetry	PostgreSQL Realtime (Change Data Capture over WebSocket)	Instant live synchronization of student test submissions directly to Teacher Copilot heatmap with zero reload.
Offline Storage	IndexedDB API (`syncManager.ts`), Client UUIDv4	Local caching of 5 curriculum concepts; offline attempt queueing with idempotent background resynchronization.

CORE ENGINEERING PRINCIPLES & SEPARATION BOUNDARY

STRICT CONSTRAINTS

1. The Architectural Firewall
The LLM is NEVER the Grader. Grading and mastery progression are 100% deterministic algorithms. The AI only provides Socratic conceptual hints.

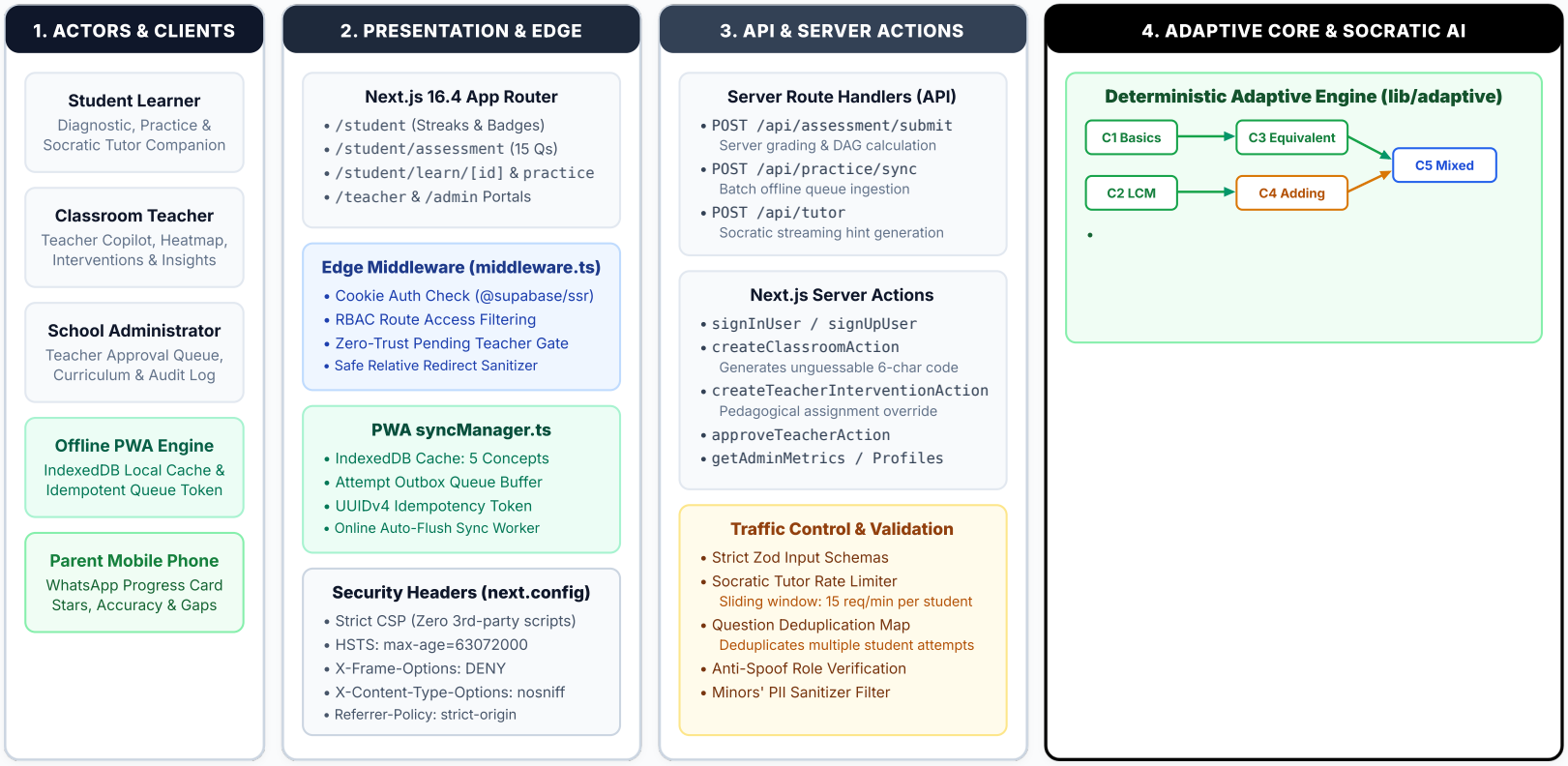
2. Zero Client-Side Grading
Clients submit raw selected options. Private answer keys remain on the server, eliminating inspection tool cheating and score tampering.

3. Minor Privacy & Isolation
Minor students' PII (names, emails) is never sent to LLM providers. Strict Row Level Security prevents cross-school data visibility.

4. Real-World Classroom Fit
Designed for erratic connectivity (IndexedDB offline PWA) and actionable teacher oversight (systemic bottleneck detection alerts).

Curriculum DAG Nodes (Class 7 Fractions):

```
C1: Fraction Basics • C2: Common Multiples & Factors (LCM) • C3: Equivalent Fractions (Req: C1)
C4: Adding Fractions (Req: C2 + C3) • C5: Mixed Fraction Problems (Req: C4)
```



Mastery Ratio Formula:
M(c) = Correct / ValidAttempts
• **Mastered:** ≥ 80% (N≥2) • **Needs Revision:** < 50%
• **Prerequisite Gap Traversal:** DFS upstream scan
• **Root Deficit Routing:** Assigns C2 before C4!
FIREWALL: LLM NEVER GRADES Socratic AI Companion (lib/ai) • **Provider Abstraction:** Gemini 2.0 Flash / OpenAI • **Indian Cultural Analogies:** Sharing rotis, dividing dosas, cricket over run-rates • **Strict Socratic Pedagogical Guardrails:** Asks probing questions; refuses answer spoon-feeding • **Zero Answer Key Leakage:** Private keys excluded from AI prompt context • **Zero Student PII Transmission:** Names, emails, and school IDs withheld from prompts • **Multilingual:** English, Hindi & Marathi prompts • **Rate-Limited:** 15 requests per minute threshold • **Anti-Hallucination:** Never computes student scores 5. SUPABASE POSTGRESQL Relational Database Tables • public.profiles (Role, Status) • public.classrooms (Join Code) • public.classroom_students • public.student_mastery • public.student_attempts • public.teacher_interventions • public.audit_logs (Immutable) • public.question_answer_keys Protected server private keys table Row Level Security (RLS) • **Students:** View own records only • **Teachers:** View assigned classes • **Admins:** Manage approval queue • **Audit Log:** Insert only; zero update PostgreSQL Realtime CDC • WebSocket Telemetry Broadcast • postgres_changes on Mastery • Auto-syncs Teacher Heatmap Matrix • Sub-second latency without polling HTTPS IndexedDB POST JSON Actions Evaluate Prompt UPSERT WebSocket CDC Live Telemetry Stream to Teacher Dashboard

Tier 1: Client & Presentation
React 19 Server Components with Next.js 16 Turbopack. Offline resilience through IndexedDB PWA attempt queueing and single-click WhatsApp progress reports.

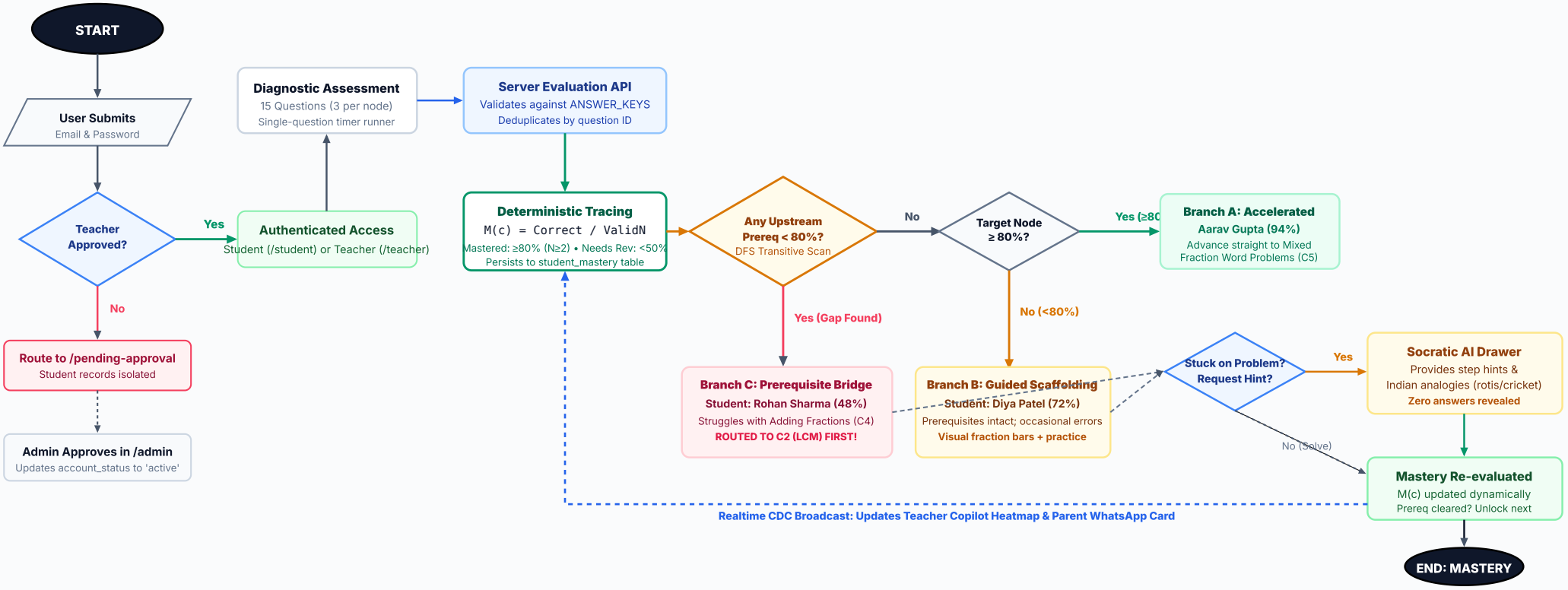
Tier 2: Edge Routing & Security
Edge middleware session cookie verification. Restricts unverified teachers to `/pending-approval`; prevents client score tampering.

Tier 3: Core Adaptive & AI Companion
Decoupled architecture: deterministic DAG knowledge tracing calculates mastery ratios; Socratic AI provides scaffolding with culturally resonant Indian analogies.

Tier 4: Relational Persistence & CDC
Supabase PostgreSQL enforces Row Level Security (RLS) on all 8 tables. Realtime CDC pushes live student mastery changes straight to the Teacher Copilot matrix.

Diagram 2: Complete ShikshaIQ Application Execution Flowchart

End-to-End Execution Sequence: Authentication, Server-Side Grading, DAG Gap Traversal, Three-Way Branching & Closed-Loop Telemetry



1. Diagnostic Baseline & Server Validation
Student initiates a 15-question baseline quiz. The client never computes score percentages; responses post to `/api/assessment/submit` where raw answers match against private server keys and deduplicate by question.

2. DFS Prerequisite Gap Traversal
The adaptive algorithm performs a topological DFS scan on target node prerequisites. If any prerequisite has `status !== 'mastered'`, it flags a
Root Gap
and routes learners to foundational bridge modules (e.g. C2 LCM before C4 Addition).

3. Socratic Scaffolding & Realtime Telemetry
When learners struggle, the Socratic AI Companion provides hints with Indian cultural analogies (rotis, dosas, cricket), refusing to reveal answers. Re-evaluation updates the Teacher Copilot matrix live via PostgreSQL CDC WebSocket.